PRODUCT INFORMATION

FLOWSIC 100

Gas Velocity and Volume Flow Measuring system

NEW

Versatile, robust, precise:

Flow Measurement in Gases with Ultrasound

The instruments of the FLOWSIC 100 range allow contact-free measurement of gas velocity and gas temperature in pipelines, ventilation and exhaust gas ducts as well as chimneys. Volume flow is determined by internal calculation and given out.

The application range extends from volume flow determination for process control and regulation to flow measurement for emission monitoring tasks according to the German regulation TA-Luft, 13th and 17th BImSchV.

Possible Applications

- Process measurements
  - Chemical industry
  - Process control throughout industry
- Industrial operating measurement and emission control, e.g.
  - Power plants
  - Waste incinerators
  - Cement and Steel Works
  - Flow measurement in ventilation, heating and air-conditioning plants

Features

- Modular concept, thereby usable for different applications
- Integral measurement independent of pressure, temperature and gas conditions
- No loss of pressure
- High accuracy and reduced sensitivity to malfunctions due digital measured value conversion
- Easy to install and operate
- Low maintenance expenditure

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Functional principle

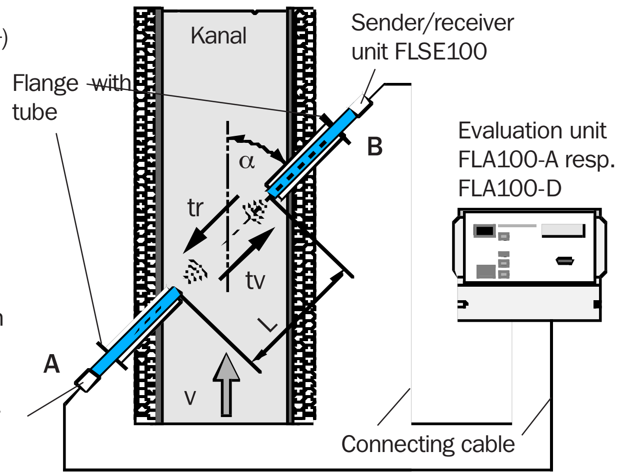
Ultrasonic transducers installed on each side of the duct act alternately sender and receiver. The accelerating and braking effects of the gas flow vary the time of flight of the ultrasonic pulses. The transit time tv is shortened in the forward direction and tr is increased in the reverse direction. The gas velocity is calculated from the transit time difference independent of pressure and temperature conditions.

FLOWSIC 100 Functional principle and Components (Cross stack)

$$v = \frac{L}{2 \cdot \cos a} \cdot \left(\frac{1}{tv} - \frac{1}{tr}\right)$$

$$Q = v \cdot A \cdot 3600$$

v… Gas velocity in m/s
L… Meas. distance in m
a… Installation angle in °
Q… Volume flow in m≈/h
A… Duct cross-section in m



Technical Data									
Measured variables	Gas velocity, volume flow in operating and normal conditions, gas temperature								
Standard measuring range	±40 m/s; continuously adjustable								
Representative accuracy	±0.1 m/s								
t_{90} time	1 ... 300 s; freely selectable								
Displays	2 line LC-display for measured variables, warning and malfunction information								
	LED for status indication operation, malfunction, maintenance and check cycle								
FLSE100 Sender/receiver unit type	PMA	PMD	PHD	UMA	UMD	UHD	USD PR	UMA PN16	UMD PN16
Standard measuring distance (m)	0,5 - 2	0,5 - 3	1 - 10	0,2 - 2	0,2 – 4*	2 - 15	0,3	0,2 – 1,4	
Inner duct diameter (m)	0,35 - 1,5	0,35 - 2,5	1,4 - 8,7	0,15 - 1,5	0,15 - 3	1,4 - 13	> 0,35	0,15 - 1	
Max. gas temperature (°C)	+ 300	+ 450		+220*			+ 200	+ 150	
Installation angle	45°... 60°						45°	45°... 60°	
Max. inner duct pressure (bar)	± 0,03**; ± 0,1***			± 0,1				16	
Distance between FLSE100 and FLA100 (m)	5	max. 1000		5	max. 1000			5	max. 1000
Output signals	analogue output 0/2/4 ... 20 mA; max. load 750 W								
	4 relay outputs 48 V, 1 A (potential free) for status signals operation/malfunction, warning, maintenance, check cycle								
Interfaces	RS 232 for parameterisation via Laptop/PC by means of MEPAFLOW program								
	RS 485 (only type FLA100-D) for data exchange between FLSE and FLA								
Options	max. 2 analogue modules 0/2/4 ... 20 m configurable as input or output								
	RS 422 interface (only FLA100-D) for communication with host computer								
	impulse output: frequency max. 200Hz / 2.5ms; power max. 40 V DC/0.5 A								
Power supply	90...140 / 180...240 V AC, 50/60 Hz, power consumption approx. 20 W								
Temperature range	-20 ... +55°C								
Protection grade	IP 65								

*: Depending on material **: with standard purge air unit ***: with purge air blower 2BH14

The Sender/receiver unit FLSE100 USD PR is designed as a probe for one-side installation.
The Sender/receiver units type FLSE100 PMA, PMD and PHD recommend a purge air unit (Option).

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